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Sequences, sums and averages

Level 2 Intermediate · Algebra

Arithmetic sequences, Gauss pairing, means handled as totals, and working backwards through a process.

By the end you can: You will be able to count and add the terms of an arithmetic sequence without fence-post errors; You will be able to turn any problem about changing means into a statement about totals; You will be able to undo a multi-step process by working backwards.

Lesson

Here is a problem to start with: twenty-five consecutive integers add up to 2025. What is the largest of them?

You could call the smallest one n and solve $n + (n+1) + \cdots + (n+24) = 2025$, and that works. But there is a faster thought. Consecutive integers are evenly spaced, so their mean is the middle one. The mean is $2025 \div 25 = 81$, so the thirteenth number is 81, and the largest is twelve more, 93. One division and one addition.

This unit is about seeing sums and averages in that way: as totals, as pairs and as balance points. We will also meet working backwards, which handles a different family of problems, where the end of a process is known and the start is not.

Arithmetic sequences

An arithmetic sequence goes up (or down) by the same amount each time. That amount is the common difference d, and the sequence is $a,\ a+d,\ a+2d, \ldots$. To reach the nth term you take $n-1$ steps, not n.

Terms of an arithmetic sequence. The nth term of the arithmetic sequence with first term a and common difference d is $a + (n-1)d$. Turned around, an arithmetic sequence that runs from a to l in steps of d has

$$\frac{l-a}{d} + 1$$

terms.

The $+1$ is the fence-post fact: a fence with 10 gaps has 11 posts. For example, the numbers from 13 to 97 that leave remainder 1 when divided by 4 are $13, 17, \ldots, 97$, and there are $\frac{97-13}{4} + 1 = 22$ of them, not 21.

Gauss pairing

To add up an arithmetic sequence, write it forwards and backwards and add the two rows column by column.

Sum of an arithmetic sequence. If an arithmetic sequence has n terms, first term a and last term l, then its sum is

$$S = \frac{n(a+l)}{2}.$$

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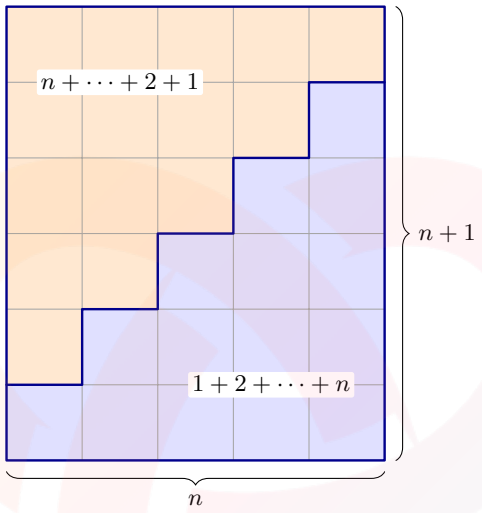
In words, the number of terms times the mean of the first and last. In particular, $1+2+\cdots+n = \frac{n(n+1)}{2}$.

Proof. Write the sum forwards and then backwards:

$$S = a + (a+d) + \cdots + (l-d) + l, \qquad S = l + (l-d) + \cdots + (a+d) + a.$$

In each column the two terms add to $a+l$, because one goes up by d exactly as the other goes down by d. There are n columns, so $2S = n(a+l)$. $\square$

For $1+2+\cdots+n$ there is also a picture. Two copies of the staircase fit together into an n by $n+1$ rectangle, so each staircase contains $\frac{n(n+1)}{2}$ squares.



The formula is also a statement about means. The mean of an arithmetic sequence is $\frac{a+l}{2}$, which is the middle term when n is odd and halfway between the two middle terms when n is even. That is what made the opening problem so quick. One more consequence worth knowing by heart: $1+3+5+\cdots+(2n-1) = \frac{n(1+2n-1)}{2} = n^2$.

Means are totals in disguise

The mean of a list is its total divided by the number of entries. Almost every problem about means becomes easy once you switch to totals, because totals simply add and means do not.

Working with means.

1. If n numbers have mean m, their total is nm.
2. If a number x is added to a list of n numbers with mean m, the new mean is
$$\frac{nm+x}{n+1} = m + \frac{x-m}{n+1}.$$
3. (Balance.) The amounts by which the numbers differ from their mean add up to zero, counting amounts below the mean as negative.

The second fact is the first one rearranged, but it is worth reading carefully: adding x moves the mean towards x by $\frac{1}{n+1}$ of the gap between them. The third is the first in another form, since

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$(x_1 - m) + \cdots + (x_n - m) = (x_1 + \cdots + x_n) - nm = 0$. It makes the mean a balance point, which is often the quickest way to think.

One warning catches people every year: the mean of two groups put together is *not* the average of their two means, unless the groups are the same size. A group of 10 with mean 50 and a group of 30 with mean 70 have combined mean $\frac{500+2100}{40} = 65$, not 60.

Working backwards

Some problems describe a process step by step, tell you how it ends and ask how it started. Going forwards means carrying an unknown through every step, and the algebra piles up. Going backwards, you undo each step with known numbers.

The rule is to undo the steps in reverse order, replacing each operation by its inverse. If a step is "halve, then subtract 1", its undoing is "add 1, then double". Order matters in the same way that you take off your shoes before your socks.

Example 1. How many terms does the sequence $7, 11, 15, \ldots, 403$ have, and what is their sum?

Solution.
The common difference is 4. A tempting quick answer for the number of terms is $\frac{403-7}{4} = 99$, but that counts the gaps between terms, not the terms. There are 99 steps of 4 from 7 to 403, so there are 100 terms. A sanity check on a tiny case helps: from 7 to 15 there are $\frac{15-7}{4} = 2$ steps and 3 terms.
The sum is the number of terms times the mean of the first and last, which is $100 \times \frac{7+403}{2} = 100 \times 205 = 20500$.
It is worth seeing the pairing directly once: $7 + 403 = 11 + 399 = 15 + 395 = 410$, and the 100 terms make 50 such pairs, giving $50 \times 410 = 20500$ again.

Example 2. A class sat a test and their mean mark was 62. Four students who were absent sat the test later. Their mean mark was 74, and once they were included the mean for the whole class rose to 64. How many students sat the test on the original day?

Solution.
There is no way to find individual marks, and you do not need any. Let n be the number of students on the original day, and convert everything to totals. The original total is $62n$, the latecomers add $4 \times 74 = 296$, and the final total is $64(n + 4)$. So $62n + 296 = 64n + 256$, which gives $2n = 40$ and $n = 20$.
The balance view does this in your head. Measure everything from the final mean of 64. The n original students are on average 2 below it, so together they are $2n$ below. The four latecomers are on average 10 above it, so together they are 40 above. Differences from the mean add up to zero, so $2n = 40$ and $n = 20$.
Notice which mean we measured from. The balance fact is about the mean of the final, combined list, so that is the one to measure from.

Example 3. A pile of counters is changed three times. Each time, one third of the counters in the pile are removed and then 4 new counters are added. After the third change there are 20 counters. How many were there at the start?

Solution.
Going forwards with x counters at the start, after one change there are $\frac{2}{3}x + 4$. After two there are $\frac{2}{3}\left(\frac{2}{3}x + 4\right) + 4 = \frac{4}{9}x + \frac{20}{3}$, and after three there are $\frac{8}{27}x + \frac{76}{9}$. Setting this equal to 20 gives $\frac{8}{27}x = \frac{104}{9}$, so $x = 39$. It works, but every line was a chance to slip.
Backwards is cleaner. One change does "multiply by $\frac{2}{3}$, then add 4". To undo it, subtract

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4 and then multiply by $\frac{3}{2}$. Starting from the end: $20 \to 16 \to 24$, then $24 \to 20 \to 30$, then $30 \to 26 \to 39$. So there were 39 counters at the start.
Always run the answer forwards as a check: $39 \to 26 \to 30$, then $30 \to 20 \to 24$, then $24 \to 16 \to 20$. The check also confirms that a third of the pile was a whole number each time, which the question quietly needed.

Example 4. In how many ways can 105 be written as a sum of two or more consecutive positive integers?

Solution.
Trial and error finds a few quickly: $52 + 53$, then $34 + 35 + 36$, then $19 + 20 + 21 + 22 + 23$. To be sure of the count, though, we need a system. Suppose the sum has k terms starting at a, so it is $a + (a + 1) + \cdots + (a + k - 1)$. By Gauss pairing this equals $\frac{k(2a+k-1)}{2}$, so we need

$$k(2a + k - 1) = 210.$$

Call the second factor $m = 2a + k - 1$. Two things are true. First, $m - k = 2a - 1$ is odd, so one of k and m is even and the other is odd. Second, $a \geq 1$ means that m is bigger than k. Conversely, any factorisation $210 = k \times m$ with k smaller than m and $m - k$ odd gives a positive whole number $a = \frac{m-k+1}{2}$, and so a valid sum.
Since $210 = 2 \times 105$ with 105 odd, in every factorisation one factor is even and the other is odd, so the parity condition is automatic here. The factor pairs with the smaller factor first are 1×210, 2×105, 3×70, 5×42, 6×35, 7×30, 10×21 and 14×15. That makes eight, but $k = 1$ is the one-term "sum" 105 itself, which is excluded. So there are 7 ways. The last pair, $k = 14$ and $m = 15$, gives $a = 1$, and indeed $1 + 2 + \cdots + 14 = 105$.
The eight factor pairs match the eight odd divisors of 105, and this is no coincidence. The same argument shows that the number of ways to write N as a sum of one or more consecutive positive integers equals the number of odd divisors of N. In particular, the powers of 2 are exactly the numbers that cannot be written as a sum of two or more consecutive positive integers.

Common traps.

- The fence-post error. From a to l in steps of d there are $\frac{l-a}{d} + 1$ terms, not $\frac{l-a}{d}$.
- Averaging averages. Combine groups through their totals, never by taking the mean of the means.
- Undoing steps in the forward order. The undoing of "add 3, then double" is "halve, then subtract 3", not "subtract 3, then halve".
- Not reading what kind of numbers are allowed. "Consecutive integers" and "consecutive positive integers" are different questions, because negative integers and zero can cancel each other out.
- Assuming the middle of an arithmetic sequence with an even number of terms is one of the terms. With an even count, the mean sits halfway between the two middle terms.

How to spot this in a contest

The triggers are evenly spaced numbers, the word "consecutive", phrases like "each time it increases by", and any question where a mean changes. With means, write down totals straight away. With consecutive numbers, think of the middle term, because the sum is the number of terms times the middle. When a problem describes a process and tells you the result, ask whether each step can be

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undone, and if it can, run it backwards. When a sum looks too long to add, look for the pairing: first with last, second with second last, and so on.

Many IMC and AMC 10 questions on this material take under a minute once you see them this way, and that buys you time for the hard questions at the end of the paper.

Practice

Try each problem before turning to the hints and solutions at the back.

1. ⋆ What is the sum of all the multiples of 3 between 100 and 200?

2. ⋆⋆ Six friends share a bag of sweets. The first friend takes half of the sweets in the bag and then one more. The second friend then takes half of the sweets that remain and then one more, and so on: each friend in turn takes half of what is left and then one more. After the sixth friend has had a turn, the bag is empty. How many sweets did the third friend take?

3. ⋆⋆ A list of n numbers has the following property. Removing its largest number lowers the mean by 2, while removing its smallest number raises the mean by 3. The largest and smallest numbers differ by 75. Find n.

4. ⋆⋆⋆ What is the largest number of consecutive positive integers that add up to 2000?

5. ⋆⋆⋆ An arithmetic sequence has an odd number of terms. The terms in the odd positions (first, third, fifth, and so on) add up to 520, and the terms in the even positions add up to 480. How many terms does the sequence have?

6. ⋆⋆⋆⋆ How many increasing arithmetic sequences of positive integers, with at least three terms, have terms that add up to 100?

Contest ladder

Real past problems that use this idea, easiest first. Open them on the official paper or the AoPS wiki, or online at kevinvariant08.github.io.

1. AMC 10A 2021 Problem 4 ⋆
2. AMC 10A 2020 Problem 8 ⋆
3. UKMT IMC 2016 Question 13 ⋆⋆
4. AMC 10A 2016 Problem 8 ⋆⋆
5. UKMT IMC 2017 Question 7 ⋆⋆
6. AMC 10A 2021 Problem 5 ⋆⋆
7. AMC 10A 2019 Problem 5 ⋆⋆
8. AMC 10B 2019 Problem 13 ⋆⋆⋆
9. UKMT IMC 2018 Question 24 ⋆⋆⋆
10. UKMT Cayley 2018 Question C5 ⋆⋆⋆
11. AMC 10A 2019 Problem 15 ⋆⋆⋆
12. AMC 10A 2021 Problem 22 ⋆⋆⋆⋆

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Hints

1. Find the first and last multiples, count the terms carefully, then pair them up.
2. Undo one turn: if r sweets are left after a friend's turn, how many were there before it?
3. Let the mean be m and write each removal as an equation between totals. Express the largest and smallest numbers in terms of m and n.
4. If there are k terms starting at a, then $k(2a+k-1)=4000$. What can you say about the parities of the two factors, and which is bigger?
5. The terms in odd positions form an arithmetic sequence of their own, and so do the terms in even positions. What is the mean of each?
6. With n terms, first term a and difference d, the condition is $n(2a+(n-1)d)=200$. First show that n cannot be large, then deal with each possible n separately.

Solutions

Problem 1

Answer: 4950

The smallest multiple of 3 above 100 is 102 and the largest below 200 is 198. They form an arithmetic sequence with common difference 3, so there are $\frac{198-102}{3}+1=33$ terms. The sum is the number of terms times the mean of the first and last, which is $33\times\frac{102+198}{2}=33\times150=4950$.

Problem 2

Answer: 16

If a friend finds y sweets, they take $\frac{y}{2}+1$ and leave $\frac{y}{2}-1$. So if r sweets are left after a turn, there were $2(r+1)$ before it. Working backwards from the empty bag, there were $2(0+1)=2$ sweets before the sixth turn, $2(2+1)=6$ before the fifth, $2(6+1)=14$ before the fourth and $2(14+1)=30$ before the third. The third friend found 30 sweets and took $15+1=16$. As a check, the bag started with $2(30+1)=62$ before the second turn and $2(62+1)=126$ at the very start, and running forwards $126\to62\to30\to14\to6\to2\to0$ works.

Problem 3

Answer: 16

Let the mean be m, so the total is nm. Let the largest number be L and the smallest be S. Removing L leaves $n-1$ numbers with mean $m-2$, so $nm-L=(n-1)(m-2)$, which simplifies to $L=m+2(n-1)$. Removing S leaves $n-1$ numbers with mean $m+3$, so $nm-S=(n-1)(m+3)$, which simplifies to $S=m-3(n-1)$. Subtracting, $L-S=5(n-1)=75$, so $n=16$.

Problem 4

Answer: 32

If there are k terms starting at a, Gauss pairing gives $\frac{k(2a+k-1)}{2}=2000$, so $k\times m=4000$ where $m=2a+k-1$. Here $m-k=2a-1$ is odd, so one of k, m is odd and the other even, and $a\geq1$ means m is bigger than k. Conversely any such factorisation gives a positive integer $a=\frac{m-k+1}{2}$.

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Since $4000 = 2^5 \times 5^3$, the odd factor must divide 125 and the even factor must contain all of 2^5. The possible factorisations are 1×4000, 5×800, 25×160 and 125×32. In each, k is the smaller factor, so k can be 1, 5, 25 or 32. The largest is $k = 32$, with $m = 125$ and $a = \frac{125-32+1}{2} = 47$. Indeed $47 + 48 + \cdots + 78 = 32 \times \frac{47+78}{2} = 16 \times 125 = 2000$. Notice that the best k here is the even factor, not the odd one.

Problem 5

Answer: 25

Say there are $2n + 1$ terms, and let M be the middle term, in position $n + 1$. The $n + 1$ terms in odd positions form an arithmetic sequence whose first and last terms are the first and last terms of the whole sequence, so their mean is the mean of those two, which is M. The n terms in even positions also form an arithmetic sequence, with first and last terms in positions 2 and $2n$, which are symmetric about the middle, so their mean is also M. Hence $(n + 1)M = 520$ and $nM = 480$. Subtracting gives $M = 40$, so $n = 12$ and the sequence has $2n + 1 = 25$ terms.

Problem 6

Answer: 20

Let the sequence have $n \geq 3$ terms, first term $a \geq 1$ and common difference $d \geq 1$. Its sum is $\frac{n(2a+(n-1)d)}{2}$, so we need $n \times m = 200$ where $m = 2a + (n - 1)d$. In particular n divides 200. Also $m \geq 2 + (n - 1) = n + 1$, so $n(n + 1) \leq 200$, which forces $n \leq 13$. The divisors of 200 from 3 to 13 are 4, 5, 8 and 10.

If $n = 4$, then $2a + 3d = 50$, so d is even, and $a = \frac{50-3d}{2} \geq 1$ needs $d \leq 16$. That gives $d = 2, 4, \ldots, 16$, which is 8 sequences. If $n = 5$, then $2a + 4d = 40$, so $a = 20 - 2d \geq 1$, giving $d = 1, 2, \ldots, 9$, which is 9 sequences. If $n = 8$, then $2a + 7d = 25$, so d is odd, and $d = 1$ gives $a = 9$, $d = 3$ gives $a = 2$, while $d \geq 5$ makes a negative: 2 sequences. If $n = 10$, then $2a + 9d = 20$, so d is even, and only $d = 2$, $a = 1$ works: 1 sequence. The total is $8 + 9 + 2 + 1 = 20$.

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